

Antifouling coatings based on the synergistic action of biogenic antimicrobial agents and low surface energy silicone resins and their application to marine aquaculture nets

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ABSTRACT

Aquaculture nets used in seawater for a long time are susceptible to biofouling, and antifouling coatings are a cost-effective way to solve this problem. In this study, biogenic antimicrobial components were extracted from two plants, Houttuynia and Scutellaria, and modified with methylphenyl silicone resin (MSR) by physical blending. Two flexible antifouling coatings with synergistic action of biogenic antimicrobial agents and low surface energy silicone resin for marine aquaculture nets were prepared separately. The prepared coatings retained the low surface energy properties of silicone resin: the contact angles (CA) were all over 130°, and the surface energy (SE) were all below 1.5 mN/m. Meanwhile, the coatings with Houttuynia and Scutellaria additives showed inhibition rates of more than 90 % against *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*), and more than 50 % against *Chlorella vulgaris*. Marine Field Test showed that the coatings containing Scutellaria and Houttuynia additives reduced fouling adhesion by 17.24 % and 35.92 %, respectively, compared with Ultra-high molecular weight polyethylene (UHMWPE) net coats without any additives. This study provides an efficient and environmentally friendly solution for antifouling treatment of marine aquaculture nets, which is of positive significance for enhancing the sustainable development of Marine Ranching.

1. Introduction

According to the Food and Agriculture Organization of the United Nations (FAO), the share of global mariculture in total fisheries production has been increasing year by year. In 2022, global aquaculture production reached 87 million tons, accounting for 49.23 % of total global fisheries production. By 2030, global aquaculture production is expected to reach 106 million tons [1]. Deep-sea development of mariculture aims to relieve pressure on near-shore resources, protect the ecosystem, improve production efficiency, meet market demand and enhance international cooperation and competitiveness [2]. However, with its increasing scale, the mariculture industry is facing more and more challenges, one of which is the problem of biofouling of aquaculture nets [3,4].

Biofouling is the attachment of animals, plants, micro-organisms and their metabolites, together with sediment and other impurities, to ships, farming equipment and other underwater artificial facilities in the marine environment [5]. Marine biofouling negatively affects the development and utilization of marine resources [6–8], especially on marine aquaculture nets. The hazards include clogging of mesh holes, hindering the growth and development of aquaculture organisms, increasing the resistance of aquaculture facilities, and increasing the cost of cleanup [3].

To solve this problem, several antifouling methods have been proposed, including manual or mechanical physical removal [5], metal synthetic mesh coating [9,10], antimicrobial synthetic fiber mesh coating [11,12], and coating antifouling [13–15]. Among them, the coating antifouling method has attracted attention due to its economic,

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simple and efficient features [16]. The antifouling mechanisms of antifouling coatings mainly include inhibiting biofouling by releasing antifouling agents through the coating [17] and utilizing low surface energy to make fouling organisms less likely to attach or easy to remove [18,19]. Metal ions are commonly used as antifouling agents in conventional methods, but this poses the risk of environmental pollution and ecosystem damage [9]. Therefore, efficient and environmentally friendly coatings are urgently needed to meet this challenge [20].

In recent years, antifouling agents of biological origin have received much attention due to their low toxicity and easy degradability [21,22]. Currently, a variety of bio-derived antifouling agents have been extracted from marine microorganisms [23], algae [24,25], benthic invertebrates [26–28] and terrestrial plants [29–31]. These biogenic antimicrobial agents offer new possibilities for environmentally friendly antifouling technologies [32].

Pan [33,34] et al. found that antifouling coatings prepared by combining butenolide from marine bacteria with modified polyurethane provided excellent antifouling performance for more than 3 months in marine field test. They also found that controlled release of butenolide could be achieved by modulating the length of the soft segment chain in the polyurethane. Desbois [35] et al. found that diatom extracts possessed strong antimicrobial activity against *S. aureus*, where free fatty acids had a significant effect on the enhancement of antifouling activity. In addition, chemical compositions of Rhodophyta, Phaeophyta and Chlorophyta also showed antifouling activity. Antimicrobial peptides SCAMPs from the hydrolysate of snow crab shells, and this antimicrobial peptide could affect the interaction of fouling biofilms with natural organic matter (NOM) to form a tuned SCAMPs-NOM, which could alter the physicochemical properties of the surface of water-soaked materials, leading to a reduction in cellular activity, altered cellular structure, which in turn inhibits biofilm formation and development. Camptothecin (CPT) is a cytotoxic quinoline alkaloid that inhibits DNA topoisomerases, thus further inhibiting conformational changes in the DNA topology, and has strong cytotoxicity. Hao [31] et al. investigated the degradation ability of CPT under different conditions and evaluated the environmental risk of its use as an antifouling agent for the sea antifouling performance on pearl oyster culture cages was good. Although many biogenic antifouling agents have been reported, not many of them have been used to prepare antifouling coatings and examined the antifouling efficacy in the actual sea area, even fewer of them have been applied in mariculture facilities. Only a few papers have reported the application of capsaicin [15] and its derivatives and butenolide [36–38] compounds on net coats. Therefore, more studies on the application of biogenic antifouling agents on mariculture facilities are urgently needed.

In the field of Chinese herbal medicines, Houttuynia and Scutellarin have attracted much attention due to their remarkable antimicrobial effects [39]. Houttuynia not only enhances immunity, antibacterial and antiviral properties, but also inhibits a wide range of microorganisms. Scutellarin exhibits broad-spectrum antimicrobial activity with significant inhibitory effects against a wide range of bacteria, including *S. aureus* and *E. coli*, and also possesses antiviral, anti-inflammatory, and immune-enhancing properties. These two herbs play an important role in Chinese medicine and health care. However, this is the first study to add these two herbs with significant antimicrobial effects to a coating for the prevention of biofouling. In this study, Houttuynia and Scutellarin extracts were applied in combination with a low surface energy silicone resin to realize the synergistic antifouling purpose of antimicrobial and antifouling by synergizing the antimicrobial agent with low surface energy.

2. Experimental section

2.1. Materials

Two herbs with significant antimicrobial properties were selected for

this study: Houttuynia and Scutellarin. These two herbs are widely used in China and have a long history of use in medicine.

The antimicrobial effects of Houttuynia are mainly derived from the various bioactive components it contains [40]. For example, compounds such as decanal and saffrole in Houttuynia have shown the ability to inhibit a wide range of bacteria and fungi. In addition, flavonoids contained in the plant, such as quercetin and kaempferol, have been shown to possess antimicrobial properties [41]. The mechanisms by which Houttuynia resists bacteria include disruption of cell walls or cell membranes, interference with bacterial metabolic processes, and antioxidant effects [42].

Scutellarin exhibits favorable antibacterial, antiviral, anti-inflammatory, and antioxidant effects [43], in which baicalin, as its active ingredient, possesses significant antiviral, anti-inflammatory, and antioxidant activities. The antibacterial mechanism of Scutellarin mainly involves disrupting the integrity of bacterial cell membranes, inhibiting the formation of bacterial biofilms, interfering with bacterial DNA replication and affecting bacterial metabolic pathways [44].

Houttuynia and Scutellarin used in this study were sourced from the local market, while the MSR used as the coating substrate was supplied by Hubei Longsheng Sihai New Material Co., Ltd.

2.2. Extraction preparation

The Houttuynia used in this experiment was freshly harvested stem and leaf parts. Houttuynia was first cleaned to remove surface impurities. The cleaned Houttuynia was placed in an oven at a temperature set at 60 °C for drying until it was completely dry. The dried Houttuynia was processed through a pulverizer to make a dried powder.

Ethanol was used as extraction solvent and Houttuynia powder was added to the extraction bottle in the ratio of 1:10 (w/v). Ultrasonic treatment was used for 30 min to promote the release of active ingredients. After completion of ultrasonic treatment, filtration was carried out and the filtrate was retained. The filtrate was subjected to a second ultrasonic extraction and filtered in the same manner. Then, all the filtrates were combined and concentrated in a water bath heater at 55 °C until the desired concentration of Houttuynia extract (@Houttuynia) was obtained.

The Scutellarin plant was pulverized, extracted and concentrated in the same way as Houttuynia to obtain the final Scutellarin extract (@Scutellarin). Usually 1 kg of Houttuynia and scutellarin can obtain 80–100 mg of effective extract.

2.3. Coating preparation

In the preparation of Houttuynia coating (Houttuynia/MSR) and Scutellarin coating (Scutellarin/MSR), we took the same steps as shown in Fig. 1. First, the MSR was mixed with the corresponding extracts at a mass ratio of 4:1. Then, the mixture was stirred at 4000 rpm for 30 min to ensure that the two materials could be mixed well and homogeneously.

The application of the coating is carried out using the dip-coating method. The specific steps are as follows: firstly, the mesh coat is dipped in the prepared coating to ensure that the mesh coat can fully absorb the coating material. Subsequently, the dip-coated mesh coat was removed and placed outdoors to dry for 2 h until the surface was dry. This process was repeated 3 times to ensure uniform coverage and sufficient thickness of the coating. After the final coating, the mesh coat was allowed to air dry for 24 h to fully cure the coating. The thickness of the cured coating is about 0.2 mm.

Using this method, we successfully prepared low surface energy coatings containing @Houttuynia (Houttuynia/MSR) and low surface energy coatings containing @Scutellarin (Scutellarin/MSR), respectively.

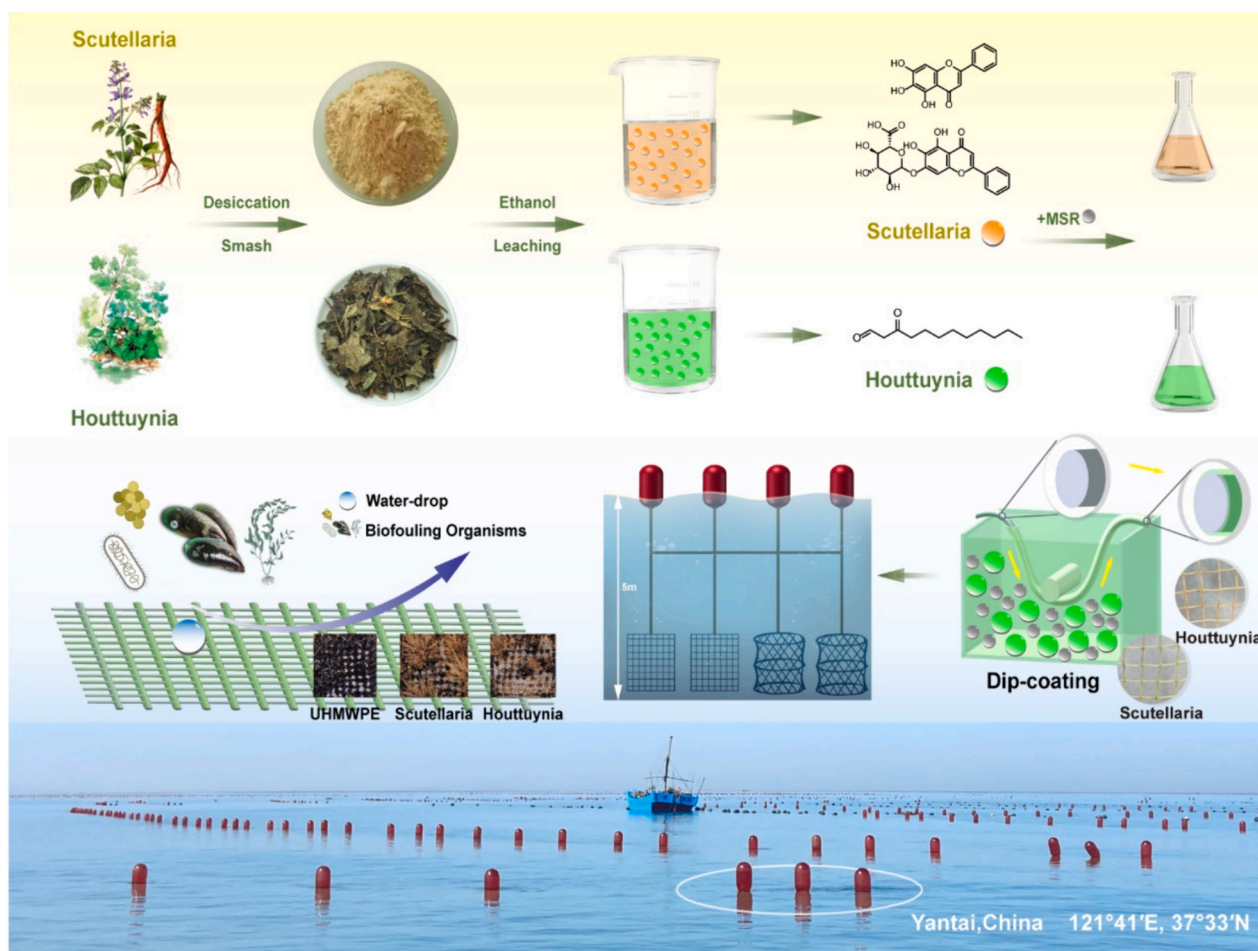


Fig. 1. The preparation process of antibacterial coatings and the state diagram of hanging the mesh.

2.4. Characteristic analysis

2.4.1. Atomic force microscopy

The film sample was fixed onto a glass slide and the surface morphology and properties were observed with a Bruker Dimension Icon AFM instrument with a scanning range of $10\ \mu\text{m} \times 10\ \mu\text{m}$.

2.4.2. Nuclear magnetic resonance spectra

A Bruker 400 MHz NMR instrument was used. In order to minimize the interference of solvent peaks, a deuterated chloroform solvent was used, and a standard pulse sequence was used for data acquisition. The carbon spectral data were sampled over a small area, and the spectra were obtained with high resolution and signal-to-noise ratio.

2.4.3. Dynamic contact angle and surface free energy

Using the KRÜSS Mechanical Method Surface Tension Meter, a coated sample was applied to a braided rope of UHMWPE fibers with a diameter of 1 mm by measuring the contact angle (CA) [45] of the sample under mediums such as deionized water and ethylene glycol. Prior to measurement, the coating was gently rinsed with deionized water to remove the effects of surface impurities and then blown dry with nitrogen. The surface free energy (SFE) was calculated according to the Owens-Wendt-Rabel-Kaelble (OWRK) method.

2.4.4. Antimicrobial and long-lasting antimicrobial capacity test

The coating was immersed in a sterile Petri dish containing 10 mL of *E. coli* suspension and *S. aureus* suspension and incubated at $37\ ^\circ\text{C}$ for 24 h before being diluted three times with PBS buffer. The last bacterial

solution of 200 μL was taken and spread on LB agar plates and incubated at $37\ ^\circ\text{C}$ for 18 h. The bacterial inhibition experiment was repeated three times to compare the antimicrobial effect of different coated samples by counting the number of colonies on the plates.

After the coating was immersed in seawater for 30 days, the above antimicrobial work was repeated, and the lasting antimicrobial ability of the coating could be observed.

2.4.5. Algae resistance and long-lasting algae resistance tests

2 mL of algal medium and the coated samples were placed in Petri dishes, which were incubated in an incubator for 24 and 72 h (12/12 dark/light cycle mode). Afterwards, the samples were rinsed three times with artificial seawater to remove the suspended unadhered algae on the surface, and water was removed from the slide surface by blotting with filter paper. Chlorella adhering to the coating was observed using an electron microscope. Fifteen photographs were taken randomly under the field of view to compare the anti-algae effect of different coated samples by counting the amount of chlorella on different fields of view [46].

After the coatings were immersed in seawater for 30 days, the above chlorella incubation was repeated to observe the long-lasting algae resistance of the coatings.

2.4.6. Marine field test

With the increase of seawater temperature, May to July is the time of the year when the growth and reproduction activities of organisms in the sea off Yantai, China, are at their peak. The marine field test were selected to be conducted from May to July in the mariculture waters of

Yantai, China (121°41'E, 37°33'N). The coatings were attached to the UHMWPE mesh coat by dip-coating, and the mesh coat pegboards (250 mm × 300 mm × 3 mm) were immersed in a water depth of 4–5 m. Two sets of parallel samples were set up for each type of mesh coat pegboard. After a period of time, the boards were removed and rinsed with seawater to remove unstable dirt and then photographed and recorded. ImageJ software was used to quantitatively analyze the contaminated area by selecting three visual fields relatively free from the frame edge in two hanging plates, and the test area only included the 30 mm × 40 mm area in the center of the mesh.

3. Results and discussion

3.1. Surface chemical composition

In order to gain a deeper understanding of the properties and mechanism of the coating, the composition and morphology of the coating were analyzed in detail [47]. In this paragraph, the performance of the coating will be analyzed in depth around the NMR spectral analysis (^1H NMR and ^{13}C NMR), Fourier Transform Infrared Spectroscopy (FTIR) and atomic force microscopy (AFM) test results of the extract components.

Fig. 2 shows the ^1H NMR and ^{13}C NMR spectra of @Houttuynia

(Fig. 2 (a) and (b)) and @Scutellarin (Fig. 2 (d) and (e)). In @Houttuynia, the major constituent 3-oxododecanal is identified through the hydrogen atom at the end of its alkyl chain at 0.9 ppm in the ^1H NMR spectrum. In addition, peaks at 1.27 ppm and 1.59 ppm correspond to hydrogen and methylene hydrogen atoms on the longer alkyl chains, respectively, while the peak at 2.30 ppm indicates the presence of hydrogen atoms attached to the carbonyl group. ^{13}C NMR spectra of 3-oxododecanal show carbon atom peaks distributed at 131.94 ppm, 127.75 ppm, 77.39 ppm, 38.61 ppm, 25.52 ppm, and 14.29 ppm, further confirming the presence of the 3-oxododecanal structure.

For @Scutellarin, ^1H NMR spectra revealed the presence of Scutellarin with characteristic peaks located at 6.58 ppm, 7.52 ppm, 7.94 ppm, and 8.44 ppm, respectively. These data indicate the presence of Scutellarin components in @Scutellarin, further validating the quality and potency of the extract.

The results of the NMR analysis not only established the presence of the major antimicrobial active components in @Houttuynia and @Scutellarin, but also provided a scientific basis for understanding the antimicrobial mechanisms of these biogenic products through detailed chemical structure analysis. These findings emphasize the value of Houttuynia and Scutellarin as potential additives for biogenic anti-fouling coatings and provide a solid foundation for future application studies.

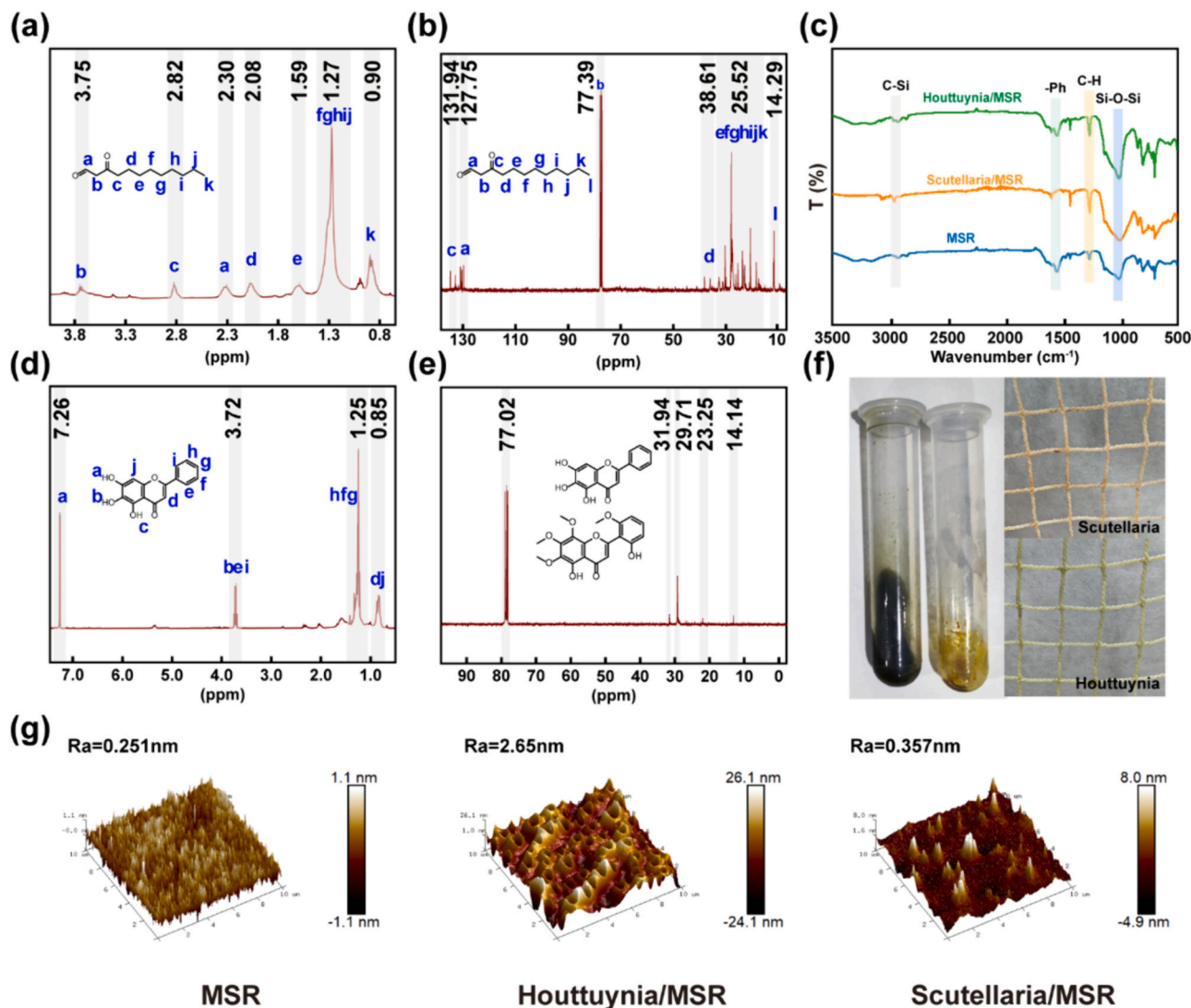


Fig. 2. Surface chemical composition: (a–b) ^1H NMR and ^{13}C NMR spectrum of @Houttuynia, (b) FTIR spectrum of coatings, (e–f) ^1H NMR and ^{13}C NMR spectrum of @Scutellarin, (f) @Houttuynia and @Scutellarin and coatings, (g) AFM image of the coated surface.

The infrared spectra of WSR, Houttuynia/WSR, and Scutellarin/WSR were analyzed using FTIR, shown in Fig. 2(c). The analytical results show that the FTIR spectra of all samples are highly consistent in terms of wave number and are in general agreement with the characteristic peaks of the control WSR.

Specifically, the peak at 2918.1 cm^{-1} is attributed to the C—Si stretching vibration of the methyl group linked to silicon, and the peak at 2848.6 cm^{-1} corresponds to the C—H bond stretching vibration of the methyl group. In addition, the peak at 1548.9 cm^{-1} represents the backbone stretching vibration of the benzene ring, the peak at 1431.4 cm^{-1} labels the bending vibration of the benzene ring, and the peak at 1262.9 cm^{-1} points to the C—H bond bending vibration of the methyl group. The stretching vibration of the Si—O—Si bond and the bending vibration of the Si—O—Si bond are manifested at 1015.8 cm^{-1} and 683.9 cm^{-1} , respectively, while the methyl group is linked to the C—Si bending vibration of the Si linkage appears at 794.3 cm^{-1} .

It is worth noting that despite the introduction of @Houttuynia and @Scutellarin as antimicrobial additives, the effect of these components on the surface structure of coating is relatively weak due to the relatively small amount of addition. This explains why the FTIR spectra of Houttuynia/WSR and Scutellarin/WSR show a high degree of similarity compared to WSR. This result indicates that the basic chemical structure of the coating remains unchanged with the addition of antimicrobial components.

The effect of the addition of antimicrobial ingredients on the surface morphology and roughness of coatings was investigated in detail in this study through AFM 3D imaging. The AFM images presented in Fig. 2(g) clearly confirmed that the addition of antimicrobial ingredients significantly affected the surface morphology of the coatings, and consequently, the magnitude of the contact angle of the material surfaces.

Specifically, the surface roughness of Houttuynia/WSR and Scutellarin/WSR was 0.521 nm and 2.65 nm , respectively, while the surface roughness of WSR without the addition of antimicrobial ingredient was

0.357 nm . Despite the differences in the roughness, these are relatively small on the nanoscale. However, even small variations in roughness had a significant effect on the contact angle, suggesting the critical role of the nanoscale surface roughness distribution morphology in modulating the hydrophilic and hydrophobic properties of the coating surfaces.

The surface morphologies of the three coatings show a uniform distribution in the AFM images, implying that they maintain consistency in contact angle performance despite the differences in roughness. This uniform distribution of surface morphology is crucial for understanding and predicting the contact angle behavior of material surfaces, as the contact angle is directly related to the hydrophilic or hydrophobic properties of the material.

Through in-depth analysis of the properties of these coated surfaces, this study not only reveals the mechanism of surface roughness on contact angle, but also provides a basis for the design and development of new coating materials with optimized surface properties.

3.2. Surface properties of coatings

The variation of contact angle under deionized water (WCA) and ethylene glycol (ECA) as test media was investigated by surface tension analysis of UHMWPE fiber coating using KRÜSS mechanistic method surface tension meter, shown in Fig. 3 (a). When deionized water was used for surface tension testing, the response of the tensiometer was triggered by the contact between the fiber and the water/air interface. The test was conducted at room temperature and the detection sensitivity was adjusted to the minimum value to ensure that the measurement was accurately triggered before the fibers were immersed in the liquid. In the experiments, each fiber was repeatedly immersed and extracted in the liquid at a uniform rate three times to obtain a series of dynamic forward and backward contact angle measurements. To avoid sample contamination, the fibers were cleaned after each measurement

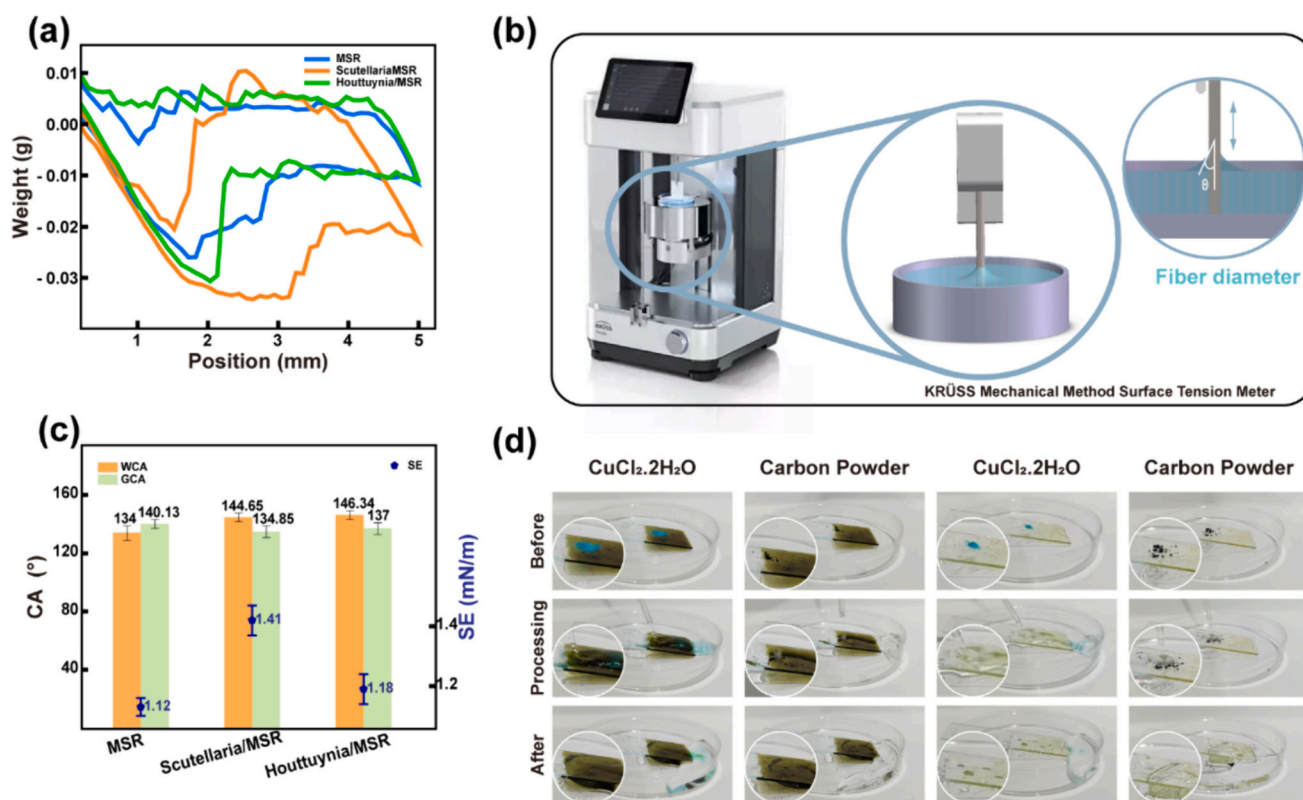


Fig. 3. Surface properties of coatings: (a) force measurement curve in water, (b) CA test equipment, (c) WCA and ECA and SE of coatings, (d) self-cleaning effect of coatings.

and rinsed with fresh water.

Fig. 3 (b) shows a schematic diagram of the measurement equipment and its main working parts. The device calculates the dynamic contact angle of the liquid on the solid surface by recording the change of force during the insertion and withdrawal of the sample plate in the liquid.

As demonstrated in Fig. 3 (c), the results of the WCA are 134° , 144.65° , and 146.34° for WSR, Houttuynia/WSR, and Scutellaria/WSR, respectively. These data indicate that despite the addition of @Houttuynia and @Scutellarin, their effect on the low surface properties of WSR was minimal, with the dynamic contact angles all remaining above 130° and SE remaining in the low surface energy range of 1–1.5 mN/m.

These results indicate that the hydrophobicity of the surface can be enhanced by the addition of @Houttuynia and @Scutellarin without significantly altering the low surface energy properties of the raw material. This is due to the fact that on rough surfaces, the interactions of water molecules with surface atoms or molecules are weakened, resulting in difficulties for water molecules to form a tight arrangement on the surface, thus reducing hydrophilicity [21]. At the same time, the tiny grooves and bumps on rough surfaces provide more hiding places for air, resulting in an increase in the contact area of water molecules with air, leading to increased hydrophobicity [48]. This subtle modulation of surface properties demonstrates the potential of improving the surface properties of materials through biogenic extracts, providing an important theoretical basis and experimental data for the design of new coating materials with excellent hydrophobicity and antimicrobial properties.

Self-cleaning ability is a key indicator for evaluating the surface performance of coatings, especially for those coatings with low surface energy properties that are effective in resisting contaminant adhesion. In

this study, the coating's fouling resistance was evaluated by performing self-cleaning experiments using contaminants with different hydrophilic and hydrophobic properties. Hydrophobic carbon powder and hydrophilic $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ were selected as test contaminants for the experiments, and both contaminants were evenly sprinkled on the coated slides and allowed to deposit for 1 h.

As shown in Fig. 3(d), the experiment was performed by placing the slide at a 10° tilt angle and slowly rinsing the surface using water droplets to simulate the process of rainwater washout. It was observed that the water flow was able to easily wash hydrophobic contaminants such as carbon powder away from the surface, while hydrophilic contaminants dissolved in the water and slid off together with the water flow. This phenomenon clearly demonstrates the self-cleaning effect of the coating in removing different types of contaminants and proves its excellent self-cleaning ability.

This study of self-cleaning performance not only demonstrates the effectiveness of coatings in removing contaminants from surfaces by water flow, but also highlights the practical application of coatings. In this way, antifouling coatings not only reduce maintenance costs but also improve surface durability.

3.3. Antibacterial and anti-algal performance

In terms of preventing biofouling, the antimicrobial and anti-algae properties of coatings are a key concern. Because in the formation process of fouling organisms, including organic macromolecules, bacteria and diatoms are the main components of biofilm. Biofilm formation is reversible because bacteria and diatoms can be separated, and reducing biofilm formation is the key to antifouling [5].

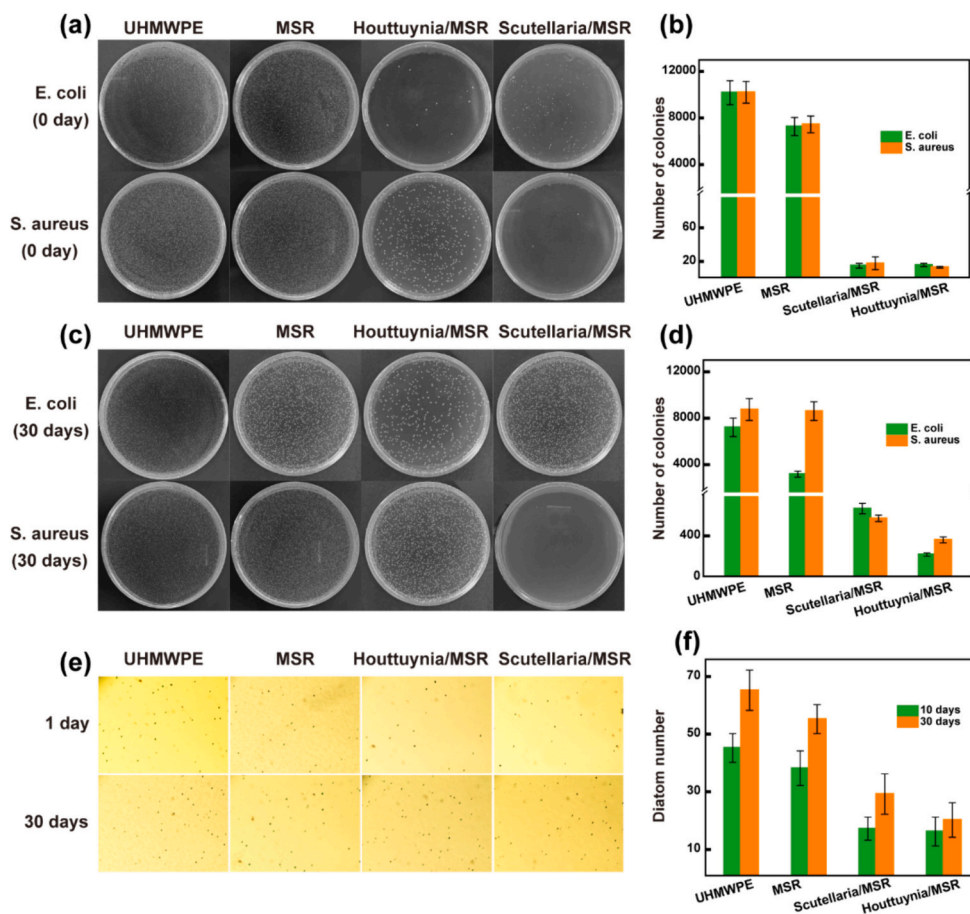


Fig. 4. (a-b) Antibacterial properties of the coating, and colony number statistics, (c-d) antibacterial properties of the coating after 30 days of immersion in seawater, and colony number statistics, (e-f) anti-algal properties of the coating, and colony number statistics.

Effective inhibition of bacterial growth and biofilm formation is one of the key strategies in the control of marine biofouling. In this study, the antimicrobial performance of Houttuynia/WSR and Scutellarin/WSR was evaluated against *S. aureus*, and *E. coli* by the plate counting method.

As shown in Fig. 4 (a), the experimental results indicated that the Houttuynia/WSR and Scutellarin/WSR coatings exhibited significant antimicrobial effects before seawater immersion. Especially, the antimicrobial efficiencies of Houttuynia/WSR coatings reached 91 % against *E. coli* and 74 % against *S. aureus*, while the antimicrobial efficiency of the Scutellarin/WSR coating achieved antibacterial efficiencies of 95 % and 99 % against *E. coli* and *S. aureus*, respectively. Because the decanoyl acetaldehyde in Houttuynia has an antibacterial effect on a variety of bacteria, including *Staphylococcus aureus* and *E. coli*. Decanoyl acetaldehyde can destroy the cell structure of bacteria and affect their growth and reproduction. Baicalein has a relatively strong bacteriostatic effect, especially on *pseudomonas aeruginosa*, *Staphylococcus aureus* and *Escherichia coli*. These data emphasize the superior performance of the two coatings with added antimicrobial agents in preventing bacterial adhesion and growth.

After 30 days of seawater immersion, the antimicrobial effect of Houttuynia/WSR, Scutellarin/WSR, as shown in Fig. 4 (c), where the antimicrobial efficiency of Houttuynia/WSR was 73.9 % for *E. coli* and 99 % for *S. aureus*. Scutellarin/WSR also had an antimicrobial rate of 91.3 % and 72.4 % against *E. coli* and *S. aureus*, respectively. The coatings still maintained a high antimicrobial capacity. This weakening effect may be due to the gradual decrease of antimicrobial components on the coating surface after long-term exposure to seawater environment. Nevertheless, both coatings demonstrated effective inhibition of bacterial growth, proving their potential for application in marine environments.

Chlorella, is widely distributed in the marine environment, has become an important indicator for studying the antifouling performance of coatings due to its strong adaptability and adhesion ability [49]. The efficacy of different coatings in resisting algae adhesion was evaluated in this study by observing the adhesion of Chlorella on the coatings through electron microscopy.

The effect of different coatings on Chlorella attachment before and after seawater immersion is shown in Fig. 4(e). Before seawater immersion, the number of algal growth on the UHMWPE plates without any coating and the MSR only plates were 44 and 37, respectively. In contrast, the addition of Houttuynia/WSR and Scutellarin/WSR showed a significant antialgae effect, with the number of algal growth decreasing to 16 and 15, respectively, which is 63.6 % and 65.9 % lower than that of UHMWPE. Indicating that these coatings are effective in resisting the adhesion of chlorella, which is the most common type of algae 65.9 %, showing that these two coatings have excellent anti-algae performance. During the experiment, it was difficult for Chlorella to attach and grow on the coated surface, and the anti-adhesion effect was very obvious compared to the uncoated control group. Houttuynia/WSR and Scutellarin/WSR had significant anti-biofouling effects, and like the results of the antimicrobial ability, the anti-algae ability of Houttuynia/WSR was also better than that of Scutellarin/WSR. The present study did not provide direct information on the inhibitory effects of Houttuynia and scutellaria on chlorella. However, given their antibacterial effects on bacteria, it can be speculated that the active ingredients in these plants may affect chlorella growth through similar mechanisms, such as by interfering with the integrity of cell membranes or inhibiting key biosynthesis pathways.

The results after 30 days of immersion in seawater further validated the same conclusion. The amount of algae growth on the uncoated UHMWPE and MSR plates increased to 64 and 54, respectively. However, the amount of algae growth on Houttuynia/WSR and Scutellarin/WSR decreased to 28 and 19, a reduction of 56.25 % and 64.81 %, respectively, which proves that the coatings maintain good algae resistance even after prolonged seawater immersion.

These results not only demonstrated the remarkable effect of Houttuynia/WSR and Scutellarin/WSR in preventing the attachment of Chlorella, but also revealed that these coatings have long-lasting anti-algal ability. Houttuynia/WSR, in particular, slightly outperformed the Scutellarin coating in terms of anti-algae performance, which is consistent with the results of its antimicrobial effect.

3.4. Marine field test

Laboratory antimicrobial and antichlorella tests to examine the antimicrobial and anti-algae properties of coatings provide a convenient method for the preliminary evaluation of antifouling performance of coatings. Laboratory experiments provide a convenient method for the preliminary evaluation of antifouling performance of coatings. Although these experiments can reflect the antifouling ability of coatings against specific organisms, the complexity of the actual marine environment far exceeds laboratory conditions. Marine environments contain thousands of species of organisms and inorganic materials that pose a comprehensive challenge to the antifouling performance of coatings. Therefore, The marine field test becomes the most direct and effective test method.

The samples illustrated in Fig. 5 were hung in seawater at a depth of about 5 m from May to July 2024 in the Yantai Sea (121°41'E, 37°33'N), China, for long-term evaluation of antifouling performance in the field. Fouling on the surface of the samples was removed by gentle rinsing with seawater to evaluate the antifouling effectiveness of different coatings against attached marine organisms.

The surfaces of the mesh coat pegboards after seawater immersion are shown in Fig. 5(a). After one month of seawater immersion, minor biofouling was observed on the surface of all samples. By analyzing the image processing of the center area (30 mm × 40 mm) of the mesh-clothes pegboards, the increase of biofouling of UHMWPE and MSR without any added coating was 7.32 % and 5.92 %. The antifouling effect of low surface energy coating is poor, which is speculated that the hanging plate is in a relatively static Marine environment, and there is not too much current shear force to carry the attached organisms off the surface. Respectively, whereas the increase of Houttuynia/WSR and Scutellarin/WSR coatings with antimicrobial agent was 1.80 % and 2.78 %, respectively, showing a significant anti-biofouling effect significant anti-biofouling effect. This was during the developmental stage of attached marine organisms, and the attachment was predominantly a mixture of marine organisms and sediment [50]. The anti-biofouling effect of the two antimicrobial coatings on the larval stage of the marine organisms was already evident.

After two months, the fouling coverage of the control UHMWPE was close to 95 %, while the coverage of MSR, Scutellarin/WSR and Houttuynia/WSR was about 88.54 %, 81.63 % and 68.00 %, respectively. Compared to UHMWPE without any additions, the amount of fouling adhesion on Scutellarin/WSR and Houttuynia/WSR was reduced by 17.24 % and 35.92 %. This indicates that Scutellarin/WSR and Houttuynia/WSR have significant advantages in resisting the biofouling of marine organisms, especially on mussels, which is significantly reduced, proving their effective antifouling performance.

In the research on the prevention and control of marine biofouling, effective inhibition of the growth and attachment of bacteria and marine organisms, especially mussels, is the key objective [51]. By comparing UHMWPE and WSR with Scutellarin/WSR and Houttuynia/WSR, this study clearly shows the advantages of the latter in inhibiting marine biofouling.

In summary, the antifouling effect on Scutellarin/WSR and Houttuynia/WSR was significantly better than WSR, which was attributed to the low surface tension of the low surface energy resins, which reduced the adhesion and diffusion of water molecules under the action of the shear force of the water flow [52]. Unlike dynamic vessels, marine aquaculture cages as static marine facilities are subjected to relatively weak flow shear. On this basis, by adding antimicrobial ingredients such as Scutellarin and Houttuynia, a gradual release of antimicrobial agents

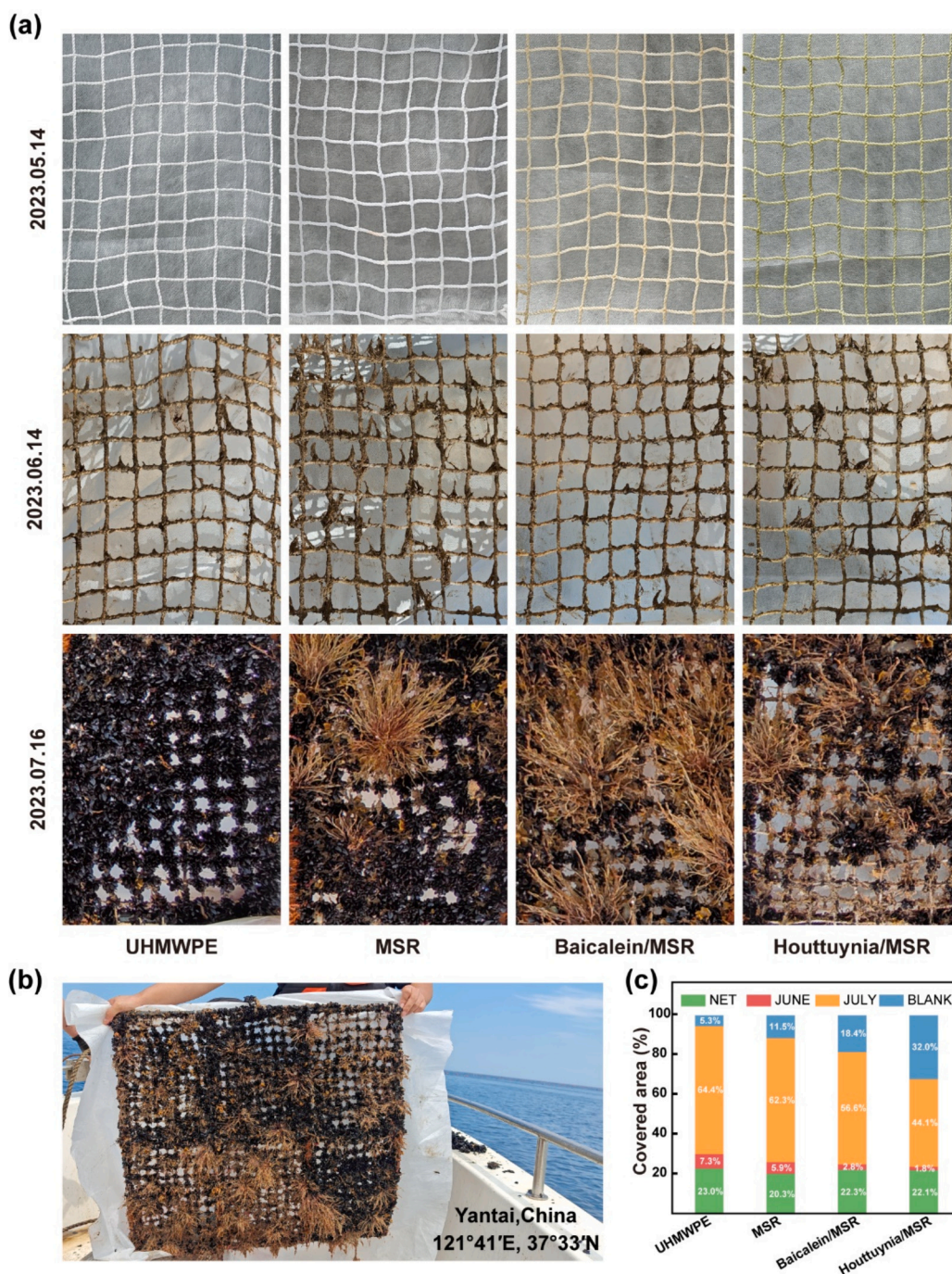


Fig. 5. Marine Field Test: (a) the effect of real sea hanging board, (b) the experiment in Yantai, (c) the coverage area of each month.

can be realized, effectively preventing the attachment and growth of marine organisms. Therefore, the development of Scutellarin/WSR and Houttuynia/WSR coatings not only demonstrates the potential for application in the field of antifouling for aquaculture nets, but also provides an important theoretical and practical basis for the design of more efficient antifouling coatings in the future.

4. Conclusion

This study analyzes the coatings with Houttuynia and Scutellarin for their antimicrobial, anti-algae, and antifouling properties through laboratory tests and marine field antifouling assessments. The experimental results lead to the following main conclusion:

Significant antimicrobial effect, the coatings containing Houttuynia exhibited significant antimicrobial effects against *S. aureus* and *E. coli* in laboratory conditions, with antimicrobial rates ranging from 91 % to 99 %. This effectively inhibited the growth of bacteria and the formation of biofilm.

Excellent anti-algae performance, the electron microscope observation and quantitative analysis revealed that the coatings added with antimicrobial agents significantly inhibited the attachment and growth of *Chlorella*. This effect was particularly evident in the anti-algae effect before and after seawater immersion, and the coatings demonstrated good long-lasting performance.

Effective antifouling performance, the evaluation of antifouling performance at the marine site revealed that Scutellarin/WSR and

Houttuynia/WSR coatings demonstrated significant advantages in reducing the number of mussels, such as mussels, adhering to the coating compared to untreated UHMWPE and MSR-only coatings. This resulted in a reduction of biofouling and mitigation of its effects.

Long-term antifouling effect, even after two months of immersion in seawater, Scutellarin/WSR and Houttuynia/WSR still maintain a low biofouling rate, demonstrating a long-term antifouling effect, particularly against mussels such as mussels.

Environmentally friendly, utilizing low surface energy and antimicrobial agents to effectively prevent the attachment and growth of marine organisms without potentially harming the marine ecology.

In summary, the development of Scutellarin/WSR and Houttuynia/WSR has provided an effective, long-lasting, and environmentally friendly antifouling coating solution. They provide an effective solution for the long-term protection and maintenance of marine aquaculture nets.

CRediT authorship contribution statement

Guifang He: Methodology, Investigation, Data curation, Formal analysis, Writing – original draft. **Hongxia Li:** Methodology, Investigation. **Zhengjie Zhao:** Investigation, Data curation. **Qingyang Liu:** Investigation. **Jingyuan Yu:** Investigation, Data curation. **Zhe Ji:** Methodology. **Xin Ning:** Methodology. **Fanggang Ning:** Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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